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Large Language Models are Zero-Shot Reasoners

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Abstract

Pretrained large language models (LLMs) are widely used in many sub-fields of natural language processing (NLP) and generally known as excellent *few-shot* learners with task-specific exemplars. Notably, chain of thought (CoT) prompting, a recent technique for eliciting complex multi-step reasoning through step-by-step answer examples, achieved the state-of-the-art performances in arithmetics and symbolic reasoning, difficult *system-2* tasks that do not follow the standard scaling laws for LLMs. While these successes are often attributed to LLMs' ability for few-shot learning, we show that LLMs are decent *zero-shot* reasoners by simply adding "Let's think step by step" before each answer. Experimental results demonstrate that our Zero-shot-CoT, using the same single prompt template, significantly outperforms zero-shot LLM performances on diverse benchmark reasoning tasks including arithmetics (MultiArith, GSM8K, AQUA-RAT, SVAMP), symbolic reasoning (Last Letter, Coin Flip), and other logical reasoning tasks (Date Understanding, Tracking Shuffled Objects), without any hand-crafted few-shot examples, e.g. increasing the accuracy on MultiArith from 17.7% to 78.7% and GSM8K from 10.4% to 40.7% with large-scale InstructGPT model (text-davinci-002), as well as similar magnitudes of improvements with another off-the-shelf large model, 540B parameter PaLM. The versatility of this single prompt across very diverse reasoning tasks hints at untapped and understudied fundamental *zero-shot* capabilities of LLMs, suggesting high-level, multi-task broad cognitive capabilities may be extracted by simple prompting. We hope our work not only serves as the minimal strongest zero-shot baseline for the challenging reasoning benchmarks, but also highlights the importance of carefully exploring and analyzing the enormous zero-shot knowledge hidden inside LLMs before crafting finetuning datasets or few-shot exemplars.

1 Introduction

Scaling up the size of language models has been key ingredients of recent revolutions in natural language processing (NLP) [Vaswani et al., 2017, Devlin et al., 2019, Raffel et al., 2020, Brown et al., 2020, Thoppilan et al., 2022, Rae et al., 2021, Chowdhery et al., 2022]. The success of large language models (LLMs) is often attributed to (in-context) few-shot or zero-shot learning. It can solve various tasks by simply conditioning the models on a few examples (few-shot) or instructions describing the task (zero-shot). The method of conditioning the language model is called "prompting" [Liu et al., 2021b], and designing prompts either manually [Schick and Schütze, 2021, Reynolds and McDonell, 2021] or automatically [Gao et al., 2021, Shin et al., 2020] has become a hot topic in NLP.

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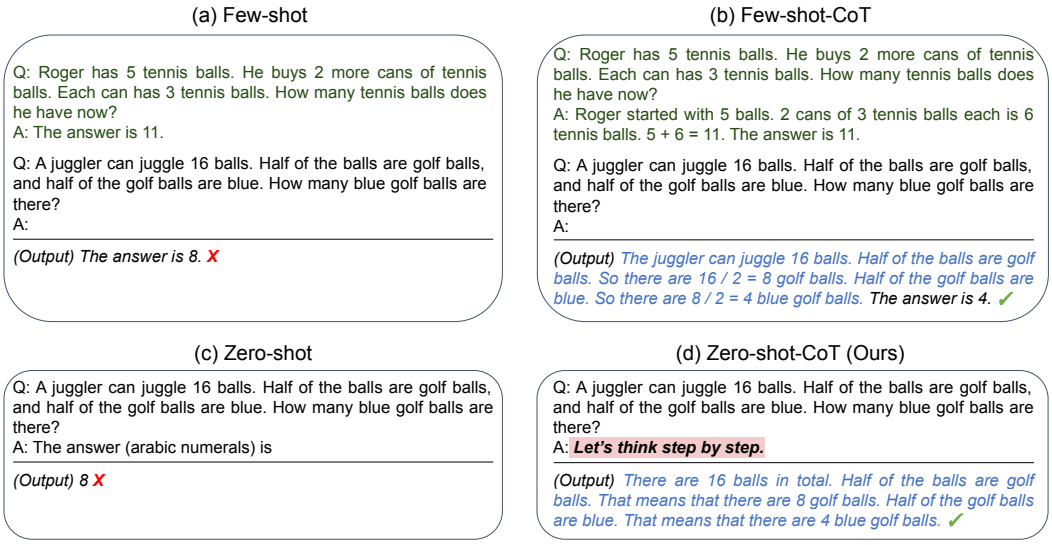

Figure 1: Example inputs and outputs of GPT-3 with (a) standard Few-shot ([Brown et al., 2020]), (b) Few-shot-CoT ([Wei et al., 2022]), (c) standard Zero-shot, and (d) ours (Zero-shot-CoT). Similar to Few-shot-CoT, Zero-shot-CoT facilitates multi-step reasoning (blue text) and reach correct answer where standard prompting fails. Unlike Few-shot-CoT using step-by-step reasoning examples **per task**, ours does not need any examples and just uses the same prompt "Let's think step by step" *across all tasks* (arithmetic, symbolic, commonsense, and other logical reasoning tasks).

In contrast to the excellent performance of LLMs in intuitive and single-step *system-1* [Stanovich and West, 2000] tasks with task-specific few-shot or zero-shot prompting [Liu et al., 2021b], even language models at the scale of 100B or more parameters had struggled on *system-2* tasks requiring slow and multi-step reasoning [Rae et al., 2021]. To address this shortcoming, Wei et al. [2022], Wang et al. [2022] have proposed *chain of thought* prompting (CoT), which feed LLMs with the step-by-step reasoning examples rather than standard question and answer examples (see Fig. 1-a). Such chain of thought demonstrations facilitate models to generate a reasoning path that decomposes the complex reasoning into multiple easier steps. Notably with CoT, the reasoning performance then satisfies the scaling laws better and jumps up with the size of the language models. For example, when combined with the 540B parameter PaLM model [Chowdhery et al., 2022], chain of thought prompting significantly increases the performance over standard few-shot prompting across several benchmark reasoning tasks, e.g., GSM8K (17.9% $\rightarrow$ 58.1%).

While the successes of CoT prompting [Wei et al., 2022], along those of many other task-specific prompting work [Gao et al., 2021, Schick and Schütze, 2021, Liu et al., 2021b], are often attributed to LLMs' ability for few-shot learning [Brown et al., 2020], we show that LLMs are decent *zero-shot* reasoners by adding a simple prompt, *Let's think step by step*, to facilitate step-by-step thinking before answering each question (see Figure 1). Despite the simplicity, our Zero-shot-CoT successfully generates a plausible reasoning path in a zero-shot manner and reaches the correct answer in a problem where the standard zero-shot approach fails. Importantly, our Zero-shot-CoT is versatile and *task-agnostic*, unlike most prior task-specific prompt engineering in the forms of examples (few-shot) or templates (zero-shot) [Liu et al., 2021b]: it can facilitate step-by-step answers across various reasoning tasks, including arithmetic (MultiArith [Roy and Roth, 2015], GSM8K [Cobbe et al., 2021], AQUA-RAT [Ling et al., 2017], and SVAMP [Patel et al., 2021]), symbolic reasoning (Last letter and Coin flip), commonsense reasoning (CommonSenseQA [Talmor et al., 2019] and Strategy QA [Geva et al., 2021]), and other logical reasoning tasks (Date understanding and Tracking Shuffled Objects from BIG-bench [Srivastava et al., 2022]) without modifying the prompt per task.

We empirically evaluate Zero-shot-CoT against other prompting baselines in Table 2. While our Zero-shot-CoT underperforms Few-shot-CoT with carefully-crafted and task-specific step-by-step examples, Zero-shot-CoT achieves enormous score gains compared to the zero-shot baseline, e.g. from 17.7% to 78.7% on MultiArith and from 10.4% to 40.7% on GSM8K with large-scale InstructGPT

model (text-davinci-002). We also evaluate Zero-shot-CoT with another off-the-shelf large model, 540B parameter PaLM, showing similar magnitudes of improvements on MultiArith and GSM8K. Importantly, with our single fixed prompt, zero-shot LLMs have a significantly better scaling curve comparable to that of the few-shot CoT baseline. We also show that besides Few-shot-CoT requiring human engineering of multi-step reasoning prompts, their performance deteriorates if prompt example question types and task question type are unmatched, suggesting high sensitivity to per-task prompt designs. In contrast, the versatility of this single prompt across diverse reasoning tasks hints at untapped and understudied *zero-shot* fundamental capabilities of LLMs, such as higher-level broad cognitive capabilities like generic logical reasoning [Chollet, 2019]. While the vibrant field of LLMs started out from the premise of excellent few-shot learners [Brown et al., 2020], we hope our work encourages more research into uncovering *high-level* and *multi-task* zero-shot capabilities hidden inside those models.

2 Background

We briefly review the two core preliminary concepts that form the basis of this work: the advent of large language models (LLMs) and prompting, and chain of thought (CoT) prompting for multi-step reasoning.

Large language models and prompting A language model (LM), is a model that looks to estimate the probability distribution over text. Recently, scaling improvements through larger model sizes (from a few million [Merity et al., 2016] to hundreds of millions [Devlin et al., 2019] to hundreds of billions [Brown et al., 2020] parameters) and larger data (e.g. webtext corpora [Gao et al., 2020]) have enabled pre-trained large language models (LLMs) to be incredibly adept at many downstream NLP tasks. Besides the classic "pre-train and fine-tune" paradigm [Liu et al., 2021b], models scaled to 100B+ parameters exhibit properties conducive to few-shot learning [Brown et al., 2020], by way of in context learning, where one can use a text or template known as a *prompt* to strongly guide the generation to output answers for desired tasks, thus beginning an era of "pre-train and prompt" [Liu et al., 2021a]. In work, we call such prompts with explicit conditioning on few task examples as *few-shot* prompts, and other template-only prompts as *zero-shot* prompts.

Chain of thought prompting Multi-step arithmetic and logical reasoning benchmarks have particularly challenged the scaling laws of large language models [Rae et al., 2021]. Chain of thought (CoT) prompting [Wei et al., 2022], an instance of few-shot prompting, proposed a simple solution by modifying the answers in few-shot examples to step-by-step answers, and achieved significant boosts in performance across these difficult benchmarks, especially when combined with very large language models like PaLM [Chowdhery et al., 2022]. The top row of Figure 1 shows standard few-shot prompting against (few-shot) CoT prompting. Notably, few-shot learning was taken as a given for tackling such difficult tasks, and the zero-shot baseline performances were not even reported in the original work [Wei et al., 2022]. To differentiate it from our method, we call Wei et al. [2022] as *Few-shot-CoT* in this work.

3 Zero-shot Chain of Thought

We propose Zero-shot-CoT, a zero-shot template-based prompting for chain of thought reasoning. It differs from the original chain of thought prompting [Wei et al., 2022] as it does not require step-by-step few-shot examples, and it differs from most of the prior template prompting [Liu et al., 2021b] as it is inherently task-agnostic and elicits multi-hop reasoning across a wide range of tasks with a single template. The core idea of our method is simple, as described in Figure 1: add *Let's think step by step*, or a a similar text (see Table 4), to extract step-by-step reasoning.

3.1 Two-stage prompting

While Zero-shot-CoT is conceptually simple, it uses prompting twice to extract both reasoning and answer, as explained in Figure 2. In contrast, the zero-shot baseline (see the bottom-left in Figure 1) already uses prompting in the form of "The answer is", to extract the answers in correct formats. Few-shot prompting, standard or CoT, avoids needing such answer-extraction prompting by explicitly designing the few-shot example answers to end in such formats (see the top-right and top-left